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Speaker recognition using Support vector machines

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SPEAKER RECOGNITION USING SUPPORT VECTOR MACHINES

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Abstract— Recently Support Vector machines are being used in a wide variety of classification purposes. This paper explores the usage of SVM's in the binary classification of speaker, as in between the original speaker and an imposter. The feature vector used for the extraction of features is MFCC's i.e. Mel Frequency Cepstrum Coefficients. The paper further explores the usage of Principle Component analysis for dimensionality reduction.

Keywords—Feature extraction, Mel frequency, Principle component analysis, Support vector machine

I. INTODUCTION

Speaker verification is a growing field and is used in a wide variety of applications such as mobile phones and a lot of bio-metric security systems. Although there have a lot of algorithms used to verify the voice of a person such as neural networks, using correlation, but they have either been very expensive for computation or not that reliable. Neural networks for example take a lot of computation time to train if they contain if they contain too many hidden layers, which they generally do if we want a good precision. Another possible way for classification would be compare with each of the training data available. This again would be very computationally costly. Support Vector Machines provide an efficient way of classification. They only require some time to train which does not exceed 20-50 iteration generally. After training the prediction on a new input data is very fast compared to neural networks. Section 2.1 of the paper gives an introduction about MFCC's i.e. about the feature vector used to extract from the signal. Section 2.2 of the paper explains principle component analysis and why is it used. Section 2.3 explores SVM's and their role in classification. In Section 2.4 we discuss the final algorithm used in the form of flowchart. Section 3 explains the graphical results, and Section 4 gives the conclusion.

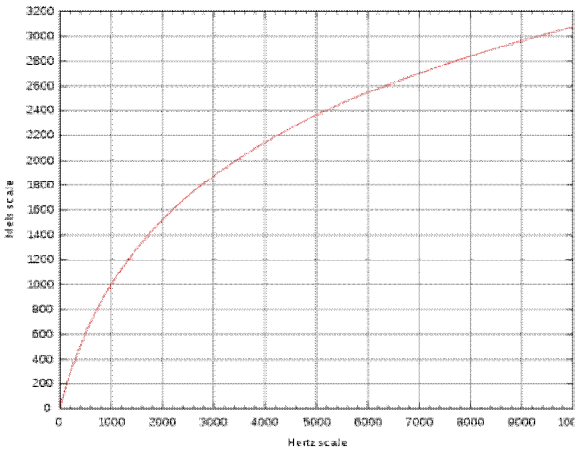
II. OUTLINE OF THE METHOD

A. Mel Frequency Cepstrum Coefficients

In sound processing, the mel-frequency cepstrum (MFC) is a representation of the short-term power spectrum of a sound, based on a linear cosine transform of a log power spectrum on a nonlinear mel scale of frequency.

Mel-frequency cepstral coefficients (MFCCs) are coefficients that collectively make up an MFC. They are derived from a type of cepstral representation of the audio clip (a nonlinear "spectrum-of-a-spectrum"). The difference between the cepstrum and the mel-frequency cepstrum is that in the MFC, the frequency bands are equally spaced on the mel scale, which approximates the human auditory system's response more closely than the linearly-spaced frequency bands used in the normal cepstrum.

a)Mel : is a scale used to measure the Pitch



The formula to convert from frequency scale to Mel is: m=2595 log(1+(f/700)) , where the log is to the base 10.

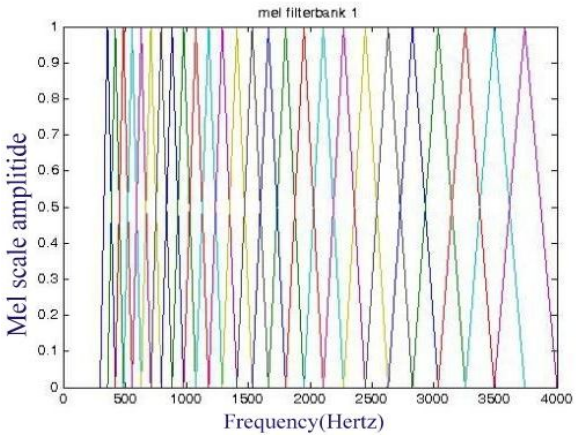
b) Cepstrum : It's the Fourier transform (FT) of the log spectrum of any signal.

B. Steps for finding the MFCC's

a. Decompose the signal into short overlapping frames, say each frame of 25 ms. Find the power spectrum of each frame.
b. Apply the Mel filter bank to each power spectra.
c. Take the log of the above energies.
d. Take the DCT (Discrete Cosine Transform) of the energies.
e. Keep the lower 12-13 DCT coefficients for each frame.

The Mel Filter bank is simply a set of triangular filters in the frequency domain. The first filter is very narrow and gives an indication of how much energy exists near 0 Hertz. As the frequencies get higher our filters get wider as we become less concerned about variations. This roughly gives how much energy occurs at each spot. The Mel scale tells us exactly

how to space our filter banks and how wide to make them.



26 Mel filter bank ranging from 300-4000 Hertz.

The Mel Filter bank gives an idea of how much energy is present in each frequency region. This is found out by multiplying the power spectrum of each frame with each Mel Filter bank

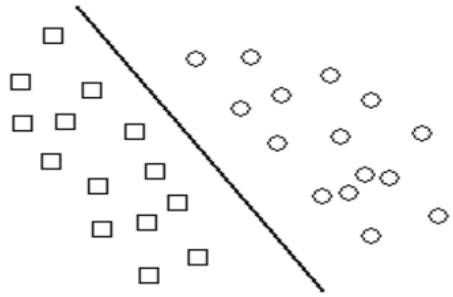
C. Principle Component Analysis

Principal component analysis (PCA) is a statistical procedure that uses orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables called principal components. The number of principal components is less than or equal to the number of original variables. This transformation is defined in such a way that the first principal component has the largest possible variance that is, accounts for as much of the variability in the data as possible, and each succeeding component in turn has the highest variance possible under the constraint that it be orthogonal to or uncorrelated with the preceding components. PCA is sensitive to the relative scaling of the original variables, so in this paper we normalized the variance of the MFCC coefficients.

PCA is most generally used for dimensionality reduction .In this paper we reduced the dimension of the MFCC coefficients from 13 to 2,by finding the most dominant 2 eigenvectors of the correlation matrix then multiplying them with the MFCC vectors.

D. Support Vector Machines

Support vector machines (SVMs, also support vector networks) are supervised learning models with associated learning algorithms that analyze data and recognize patterns, used for classification and regression analysis. The basic SVM takes a set of input data and predicts, for each given input, which of two possible classes forms the output, making it a non-probabilistic binary linear classifier. Given a set of training examples, each marked as belonging to one of two categories, an SVM training algorithm builds a model that assigns new examples into one category or the other. An SVM model is a representation of the examples as points in space, mapped so that the examples of the separate categories are divided by a clear gap that is as wide as possible. New examples are then mapped into that same space and predicted to belong to a category based on which side of the gap they fall on. In addition to performing linear classification, SVMs can efficiently perform a non-linear classification using what is called the kernel trick, implicitly mapping their inputs into high-dimensional feature spaces.

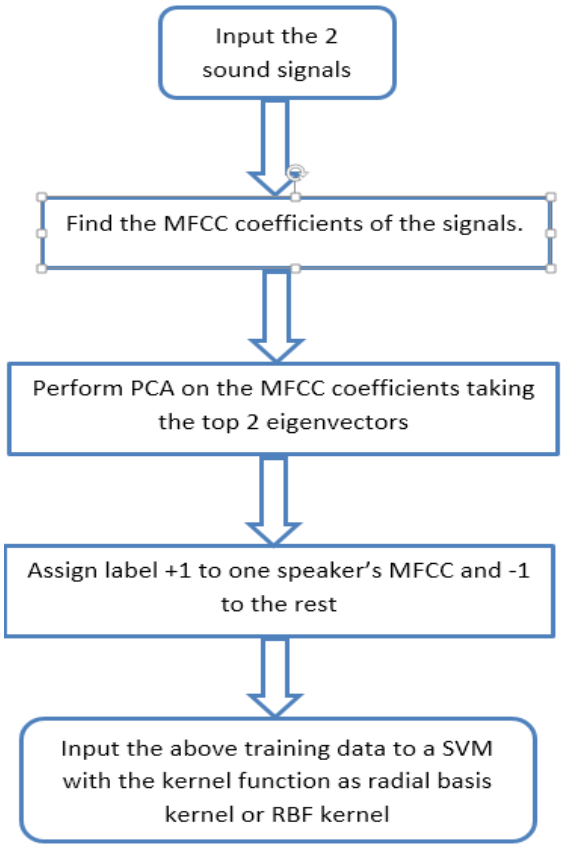


A simple SVM as a hyperplane classifier

In this paper we trained a SVM using the MFCC vector as the input feature vector and giving one speaker as label 1 and the rest as label -1 using the (Gaussian) radial basis function kernel or the RBF kernel with the sigma value being 1.

E. Final algorithm

The below given flow chart describes the entire algorithm used for speaker verification.



III. RESULTS

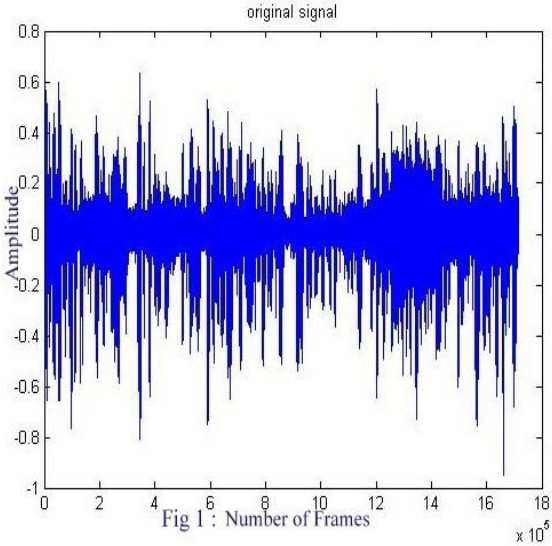


Fig 1 : Number of Frames

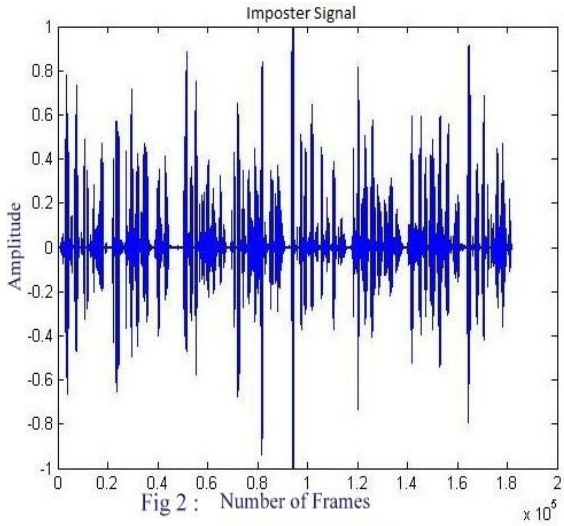


Fig 2 : Number of Frames

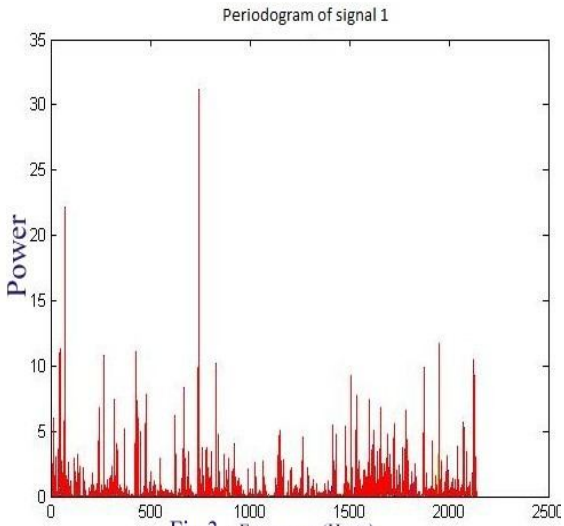


Fig 3 : Frequency (Hertz)

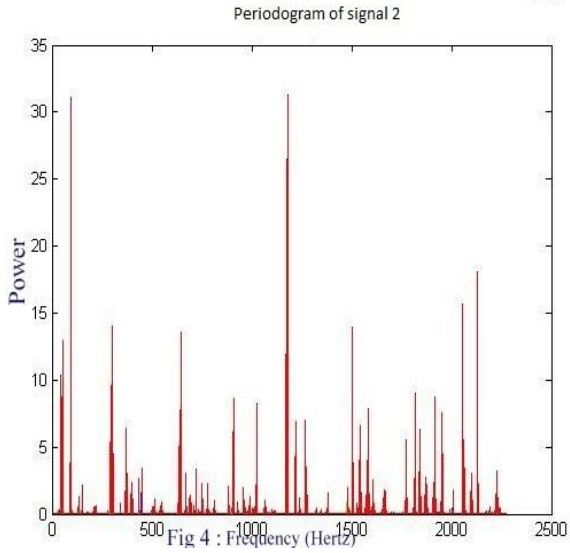


Fig 4 : Frequency (Hertz)

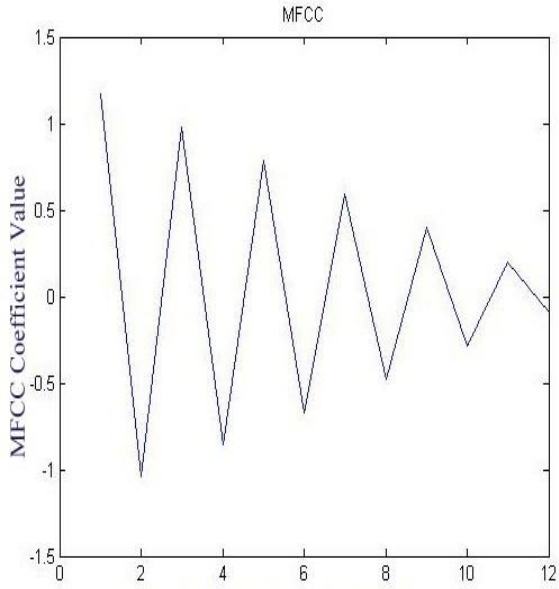


Fig.5 : MFCC Coefficient number

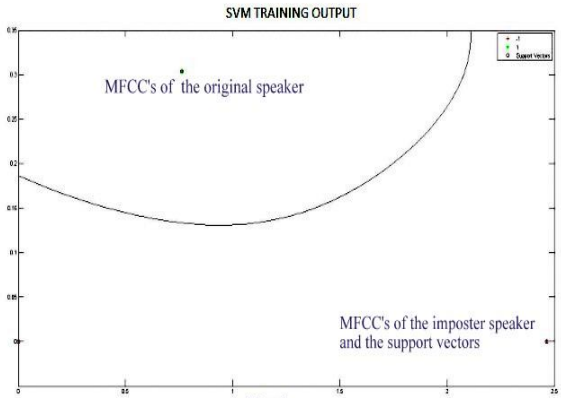


Fig.6

CONCLUSION

The paper demonstrates an efficient way of classifying the voices to 2 or more persons using Support Vector machines as the classification technique. The paper further demonstrates the use of Principle Component Analysis for the dimensionality reduction of the Mel Frequency Cepstrum coefficients from 13 dimensions to two dimensions.

The figures 1 and 2 show 2 different speakers which is further demonstrated by their periodograms in figures 3 and 4 respectively. Figure 5 shows a sample MFCC coefficient of a speaker. Fig 6 the graph of the SVM was able to separate the MFCC's coefficients of the 2 speakers by the use of the Gaussian radial basis kernel.

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